-

Cloud computing lab internal question paper

Write all the questions:

Q1. CloudSim Simulation — Arithmetic Task Scheduling (10 Marks)

Write a CloudSim program to create a simple cloud environment with the following specifications:

1. **Environment Setup**
   - Create **one Datacenter**, **one VM**, and **multiple Cloudlets**.
   - Each Cloudlet should perform a **different arithmetic operation** (e.g., Addition, Subtraction, Multiplication, Division, Sorting).
2. **Implementation**
   - Use `switch-case` statements to perform the arithmetic operations inside Cloudlets.
   - Assign different Cloudlet lengths to simulate varying execution times.
   - **Sort the Cloudlets based on their expected arithmetic execution time** before scheduling.
3. **Execution and Output**
   - Run the simulation.
   - **Display the result of each Cloudlet** after it finishes execution.
   - Example output:

```
--- Cloudlets sorted by expected arithmetic execution time ---
Cloudlet 1 (Length: 1000)
Cloudlet 2 (Length: 2000)
Cloudlet 4 (Length: 3000)
Cloudlet 3 (Length: 4000)
=========== ARITHMETIC TASK RESULTS (SORTED SCHEDULING) ===========
Cloudlet 1 completed. -> Addition Result: 15
Cloudlet 2 completed. -> Subtraction Result: 13
Cloudlet 4 completed. -> Multiplication Result: 24
Cloudlet 3 completed. -> Sorted Array: [1, 2, 5, 7, 9]
=================================================================
```

Cloud computing lab internal question paper

Write all the questions:

Q2. Set up legacy Linux inside VirtualBox.

2.1 Write a **short Python or shell script** inside `file1.txt` (minimum 5 lines).

2.2 Display system information (e.g., OS version, kernel version).

2.3 Show current logged-in user details using______ and________ commands.

2.4 Ping the **gateway** or another VM to check network connectivity.

2.5 Rename `file3.txt` to `backup.txt`.

Q3. File Transfer Between Virtual Machines

3.1 Create a file named **`labfile.txt`** on **VM1(legacy Linux)** ,(IP 10.11.22.33)and write at least **10 lines of content** (e.g., a small code snippet text) Transfer to VM2 (IP 192.0.5.5)

3.2. Display the contents of the file to confirm successful transfer.VM1

3.3. Verify that the file has been received successfully on VM2.

3.4 Set up a shared folder between two VirtualBox VMs and show the steps to mount and access it on both machines.

3.5 Describe a scenario where using a **shared folder** is better than network transfer methods.